Integration/Execution Timeline

TEXAS A&M UNIVERSITY
Engineering

	1/18/21	1/25/21	2/1/21	2/8/21	2/15/21	2/22/21	3/1/21	3/8/21	3/15/21	3/22/21	3/29/21	4/5/21	4/12/21	4/19/21	4/26/21	5/3/21	Date
Status Update 1																	1/25/21
Comms/Periph. Vib. Control Integration																	2/8/21
Power Subsystem/Sensing Unit Integration																	2/8/21
Status Update 2											Not Started						2/8/21
Signal Processing and Sensing Integration											In Progress						2/15/21
Sensing Unit Housing Design											Completed						2/15/21
Peripheral Revised PCBs Manufacturing											Behind Schedule						2/21/21
Design Blitz Participation																	2/21/21
Status Update 3																	2/22/21
Comms/SigProc Multithreading Integration																	3/1/21
Sensing Unit Housing Manufacturing																	3/1/21
Peripheral Housing Manufacturing																	3/8/21
Status Update 4																	3/8/21
Sensing Unit Mounting Design																	3/15/21
Status Update 5																	3/22/21
System Validation																	4/5/21
Final Design Presentation																	4/5/21
TI Corpate Demonstration																	4/5/21
TI Corporate Presentation																	4/8/21
TI Weekly Connectivity Podcast																	4/9/21
Final System Demo																	4/19/21
Virtual Project Showcase Video																	4/19/21
Final Report																	4/29/21

Validation Plan

TEXAS A&M UNIVERSITY Engineering

Paragraph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.1	Communication Range	Sensing unit and peripherals communicate object data and peripheral status when seperated at all ranges up to 7ft.	Peripheral self-test and object reporting functions are performed at 1ft intervals ranging from 0ft to 10ft.	UNTESTED	Evan Ross Kyle Patrick
3.2.1.2	System Latency	Haptic feedback is triggered no greater than 0.5s after object enters detection range.	Object is placed within system FOV. A stopwatch is used to time the delay between the object entering the system FOV and haptic feedback on the peripherals.	UNTESTED	Full Team
3.2.1.3	Wireless Connection Stability	Sensing unit and peripherals connection does not drop more than 1 time per 5 minutes and shall reconnect within 10s.	System is put into default operational state (5ft peripheral/sensing unit sep.) and left to run for 30min. Connection stability is monitored via UART connection on peripherals.	UNTESTED	Evan Ross
3.2.1.4	Operation Time	System operates continuously on battery power for a minimum of 1 hour.	System is put into default operational state (5ft peripheral/sensing unit sep.) and left to run for 60min. Power is monitored via UART connection with peripherals.	UNTESTED	Evan Ross
3.2.1.5	Field of View (FOV)	System detects objects with 60 degrees radially w.r.t direction user is facing.	Measure the system's response to objects that will be placed inside and outside of the target field of view	UNTESTED	Will Pinon
3.2.1.6	Detection Range	System detects objects 6ft radially from device with specified FOV.	Measure the system's response to objects that will be placed inside and outside of the target range	UNTESTED	Will Pinon
3.2.1.7	Detection Accuracy	System reports false detection less than 5% of operation time.	Place objects in detection range across 30min duration. Monitor detected objects over UART on sensing unit. Compare results.	UNTESTED	Will Pinon Evan Ross
3.2.1.8	Detection Sensitivity	Objects with angular resolutions as small as 2deg are detected.	Calculate resolution of object and determine if it is detected	UNTESTED	Will Pinon Evan Ross
3.2.1.9	Ease of Use	Sysem is attached to the user, by the user, in less than 3 minutes.	Use stopwatch while user is putting on device	UNTESTED	Kyle Patrick, Evan Ross
3.2.2.1	Mass	Mass of the sensor and signal processing unit shall not exceed 1lb	Measure sensor and signal processing unit with a digital scale	UNTESTED	Kevin Kronenberger
3.2.2.2	Volume Envelope	The sensor and signal processing unit of the system shall be contained in an enclosure with an envelope of no more than 100 cubic inches. The volume envelope of a single peripheral will not exceed 2.5 cubic inches.	Perform area measurements on the enclosures created for the sensing unit and the peripheral devices	UNTESTED	Full Team
3.2.2.3	Mounting	The sensing unit is able to be mounted to the user's head or torso. The peripheral units are able to be mountable to the user's arms and legs.	Calculate the center of mass of the sensing system and determine if objects can be mounted on user	UNTESTED	Full Team
3.2.3.1.1	Input Voltage (Sensing Unit)	The input voltage level for the TI mmWAVEICBOOST board shall be 5V and the input voltage level for the TI CC2652 Launchpad device shall also be 5V.	Use multimeter to validate input voltage levels	UNTESTED	Kevin Kronenberger
3.2.3.1.2	Input Voltage (Peripherals)	The input voltage level for the peripheral devices shall be 2.7V – 4.3V. The power consumption shall not exceed 0.5W.	Use multimeter to validate input voltage levels	UNTESTED	Kyle Patrick
3.2.3.1.3	Peripheral Battery Charging Voltage and Current	Each peripheraluses a single 3.7V 150mAh Li-Ion battery as a power supply. These batteries are able to be charged by an external USB (5V input) charger that can supply 100mA of charge current with a 4.2V charge voltage.	Use multimeter and oscilloscope to validate voltage levels and charge current levels	UNTESTED	Kyle Patrick
3.2.4.1	Thermal Resistance	The system should be able to operate in evinronments with temperatures raning from 0C to 45C	Use heating device (oven?) to raise temperature to 45C and then test system functionality. Place system inside of freezer set to 0C and test it once it has soaked there.	UNTESTED	Evan Ross
N/A	Full System Demo	A user of the system is able to navigate an obstacle course and an environemnt they are unfamiliar with.	Team member wears the full system and an obstacle couse is created that they must pass through. This person could also be lead outside and told to navigate a certain distance.	UNTESTED	Full Team